DSA REFERENCE

# Prefix Sum Dry-Run Workbook

Trace tables and blank workspaces for deliberate practice.

Precompute cumulative totals so any range aggregate becomes a difference of two prefix states.

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DSA / PREFIX-SUM

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# Prefix Sum Dry-Run Workbook

**CORE IDEA**

**Precompute cumulative totals so any range aggregate becomes a difference of two prefix states.**

### Recognition cues

+ many range-sum queries
+ subarray sum target
+ balance of categories
+ immutable array aggregation

### Implementation sequence

+ choose a zero-length prefix
+ define interval convention
+ combine with a frequency map for target subarrays
+ promote numeric type when sums grow

## Invariant

prefix[i] represents exactly the first i elements, making range [l, r) equal prefix[r] - prefix[l].

## Worked reasoning

A subarray ending at current prefix P sums to k for every earlier prefix P-k.

## Complexity

O(n) preprocessing and O(1) range queries, or O(n) for target-subarray counting with a map.

## Failure modes

- [ ] off-by-one ranges
- [ ] forgetting the initial zero prefix
- [ ] integer overflow
- [ ] using it for frequently updated arrays without a Fenwick tree

Practice ladder

# Prefix Sum Dry-Run Workbook: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

| # | Problem | Deliberate-practice prompt |
|---|---|---|
| 1 | **Range Sum Query - Immutable** | State the cue, invariant, edge case, and proof before implementation. |
| 2 | **Subarray Sum Equals K** | State the cue, invariant, edge case, and proof before implementation. |
| 3 | **Contiguous Array** | State the cue, invariant, edge case, and proof before implementation. |
| 4 | **Product of Array Except Self** | State the cue, invariant, edge case, and proof before implementation. |
| 5 | **Continuous Subarray Sum** | State the cue, invariant, edge case, and proof before implementation. |

## Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

## Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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